# A Bi-Prototype BDC Metric Network With Lightweight Adaptive Task Attention for Few-Shot Fine-Grained Ship Classification in Remote Sensing Images

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***Abstract*— Fine-grained ship classification in optical remote sensing images is a major challenge in the ocean observation field, elaborated as follows: first, the cost of acquiring ship images is expensive. Obtaining numerous labeled samples is difficult, resulting in the poor generalization ability of training models. Second, the features of ship target cannot be accurately obtained owing to complex background interference. Third, interclass similarity and intraclass diversity among different ships render ship classification difficult. In this study, we propose lightweight adaptive task attention-bi-prototype-Brownian distance covariance (LATA-BP-BDC): a BP-BDC metric network with LATA for few-shot fine-grained ship classification. First, the LATA module is used to generate 3-D weights, which can effectively reduce complex background interference and improve the adaptive capturing ability of target features without including additional network operators. Second, we input target features into the BDC metric module and output the BDC matrices to represent image information. Because the similarity between two images can be calculated as the corresponding BDC matrices distance, the improvement of the relevance of similar targets can be realized. Finally, we use the bi-prototype module to generate highly accurate prototypes, further calibrating information differences between images, which enhances the correlation between the same category samples and separability between different categories samples. Consequently, this process effectively reduces the influence of large intraclass appearance variation and small interclass appearance variation. We perform validation using two fine-grained datasets, fine-grained ship classification in remote sensing images (FGSCR), and caltech-ucsd birds-200-2011 (CUB). Compared with the state-of-the-art methods, the LATA-BP-BDC achieves a superior performance and has good generalization for fine-grained few-shot classification.**

***Index Terms*— Bi-prototype, brownian distance covariance (BDC) metric, few-shot learning (FSL), lightweight adaptive task attention (LATA), ship classification.**

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## I. Introduction

With the integration of the global economy, measuring and monitoring human activities at sea (e.g., commercial trade, port construction, maritime transport and security, and anti-illegal measures) have become increasingly important. Fine-grained ship classification of optical remote sensing images is a basic technique for maritime monitoring and has widespread applications both in civil and military domains [1], [2]. In recent years, fine-grained visual classification (FGVC) based on deep learning has achieved great success [3], [4], [5], but learning models involve abundant annotated data and complex iterations to train numerous feature parameters. However, the cost of acquiring ship images tends to be high, and for some types of ships, the number of images is relatively sparse because of their special characteristics, causing difficulties in acquiring large amounts of annotated data. Thus, the few-shot learning (FSL) method has become a new hot spot of research in recent years and only requires the use of a few labeled samples to classify new images.

At present, FSL methods are mainly grouped into three categories: data-augmentation-based [6], [7], [8], [9], [10], [11], [12], [13], [14], optimization-based [15], [16], [17], [18], [19], [20], [21], [22], [23], and metric-based methods [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41], [42]. Data-augmentation-based methods primarily synthesize features by mining the potential attributes or semantic information of objects in images or by directly expanding few-shot datasets using image generation techniques to alleviate the overfitting problem arising from insufficient data for FSL. These methods have relatively easy methodologies. However, the difficulty lies largely in avoiding the introduction of enormous amounts of noise into the augmented data, which causes interference in the training of the network. The core of optimization-based methods is to allow the model learn to fine-tune. This class

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of methods is divided into two primary routes. The first one is learning a good model initialization through auxiliary data. The second route is learning a good optimizer and replacing the existing one with the newly learned one. This model can be better accustomed to new few-shot assignments. Overall, optimization-based methods suffer from some domain offset phenomena, and their network designs do not match well for FGVC. Metric-based approaches first learn features using a feature extractor, then calculate the distance between query image features, and support image features using a metric classifier, and they use the closest sample class as their recognition result. For instance, Wang et al. [24] propose a hybrid inference network (HIN), and the samples in the embedding space are classified by combining the way of the inductive and transductive inference. Yang et al. [25], [26] propose a mixed loss graph attention network (MGA-Net), the relationships between samples are captured by the attention mechanism, and then, the classification results are obtained by updating based on a new mixed loss. Subsequently, they utilize a hierarchical embedding network (HEN) as a feature extractor to extract multiscale features and then use the center calibration algorithm to explicitly calibrate the new class centers, improving the classification performance. To summarize, the advantage of metric-based methods is that they are extremely convenient and efficient for performing new tasks, in which the model can be used directly by retaining the original parameters unchanged. Comparatively, metric-based approaches have become a more promising method for FSL for FGVC (FSL-FGVC) problems.

However, metric-based methods for ship classification still face significant challenges, and the main problems are as follows [1], [2].

1) The cost of acquiring ship images is often high, making it incredibly tough to gain abundant labeled data, which easily causes overfitting in deep neural models with many parameters.
2) As shown in Fig. 1, ship images contain relatively complex backgrounds, such as harbors and sea clutter, as well as the presence of multiple ships in one image, which causes severe interference in the extraction of target ship regions.
3) Interclass similarity [Fig. 1(a)] and intraclass diversity [Fig. 1(b)] among diverse ships also make ship classification a challenging task.

As metric-based methods are mainly composed of two parts: a feature extractor and a metric classifier, the previously mentioned challenges imply that to solve FSL-FGVC problems, first, a feature extractor should be developed that can effectively highlight important feature regions and suppress the interference of other irrelevant regions. Second, a good metric classifier is needed that can strengthen the relevance between the same category of images and the irrelevance between different categories of images. Based on these considerations, we propose a novel network for few-shot fine-grained ship classification. The main contributions of this study are as follows.

1) A bi-prototype Brownian distance covariance (BDC) metric network with lightweight adaptive task attention (LATA) for few-shot ship fine-grained classification (hereby referred to as LATA-BP-BDC) is proposed. With the currently popular meta deep BDC (Meta DeepBDC) [37] as the baseline, the LATA-BP-BDC can further enhance the fine-grained ship classification capability.
2) Compared with the baseline, a LATA module and a bi-prototype module are constructed in the feature extractor and metric classifier of LATA-BP-BDC, respectively. The former adaptively captures representative features and reduces complex background interference by assigning different 3-D weights (i.e., containing both channel and 2-D spatial dimensions) to different region descriptors. The latter utilizes positive and reverse transmission to generate highly accurate prototypes, making the distance between the same category samples smaller and that between different categories samples larger.
3) Experiments on two fine-grained datasets, fine-grained ship classification in remote sensing images (FGSCR) and caltech-ucsd birds-200-2011 (CUB), illustrate that the LATA-BP-BDC has a high-efficiency classification ability and outperforms other state-of-the-art methods.

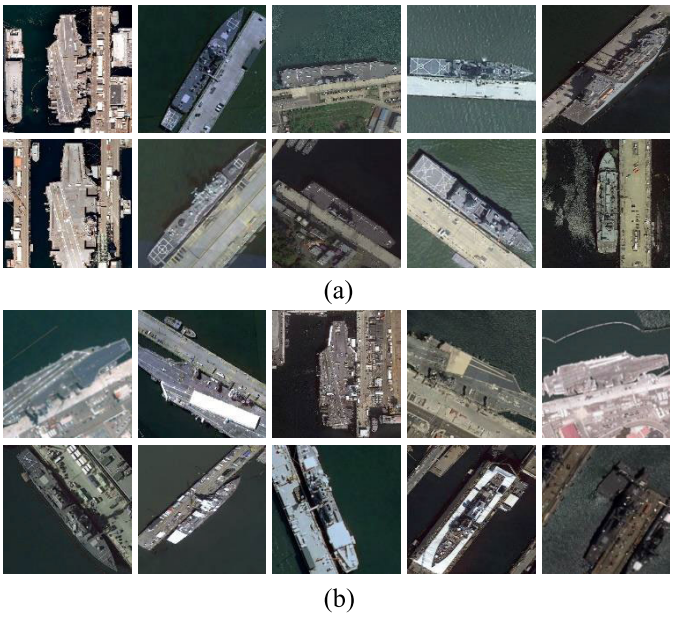


Fig. 1. Visualization of interclass similarity and intraclass diversity in ship classification. (a) Interclass similarity: Nimitz versus KittyHawk, Arleigh versus Ticonderoga, Whitby versus Antonio, Hyuga versus Lzumo, and Crane versus Sacramento (from left to right and from top to bottom). (b) Intraclass diversity: Nimitz (first row) and Arleigh (second row).

The remainder of this study is organized as follows. We briefly introduce related works in Section II. Section III presents the LATA-BP-BDC in detail. The experimental results are described and analyzed in Section IV. Section V presents the conclusions of this study.

## II. Related Work and Motivation

We briefly review related metric-based methods for FSL-FGVC and introduce them on the basis of feature extraction

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and metric classification performances, respectively, thereby leading to the creative motivation of this study.

### A. Metric-Based Methods for FSL-FGVC

*1) In Terms of Feature Extraction:* The precondition for metric-based methods in FSL-FGVC is to learn a valid embedding space. However, for fine-grained datasets, which generally contain highly complex backgrounds, this challenge can directly influence the embedding representation of different classes, potentially resulting in a misjudgment of input images by the model. Therefore, methods that can extract more representative features are required.

In this section, some popular metric-based methods with respect to feature extraction for FSL-FGVC are introduced. For example, the cross-attention network (CAN) [27] introduces a cross-attention module for every pair of class features and query image features to produce cross-attention maps to differentiate the extractive features. The position-aware relation network (PARN) [28] introduces a deformable feature extractor using deformable convolutional layers, thus efficiently extracting features. The adaptive task-aware local representations network (ATL-Net) [29] introduces a situational attention module that can select critical blocks and does not excessively notice the common parts of the entire task. The MultiScale adaptive task attention network MultiAttention deepEMD network (MAEMD-Net) [30] uses global and local attention modules to extract discriminative features from both global and local levels, respectively. A Relational Embedding Network (ReNet) [31] proposes autocorrelation representation and cross correlation attention mechanisms that work together to learn related embeddings, emphasizing target object regions. Multiscale Adaptive Task Attention Network (MATANet) [32] is a multiscale attention module that uses four average pooling layers to generate multiscale features, which can select the most discriminative local features at multiple scales for a specific task.

*2) In Terms of Metric Classification:* Metric-based methods for the FSL-FGVC measure the similarity between images. Due to the extreme scarcity of labeled data and the interclass similarity and intraclass diversity between images, image representation and similarity metrics play a crucial role in few-shot fine-grained classification [37].

Therefore, advanced metric-based methods with respect to the metric classification for FSL-FGVC are discussed as follows. The Prototypical Network (ProtoNet) [33] uses the first-order moments (i.e., mean vectors) to represent images and adopts the Euclidean distance as the similarity metric. Relation Network (RelationNet) [34] learns a transferable depth metric by designing a neural network. The Covariance Metric Network (CovaMNet) [35] constructs a local covariance to acquire the second-order statistical information of every sample and designs a covariance metric as a relational metric to calculate the distributional consistency between query samples and classes. Asymmetric Distribution Measure Network (ADM) [36] utilizes an asymmetric metric (i.e., the Kullback-Leibler (KL) divergence measure to represent the global relationship and a combination of the mean vector and covariance matrix measure to represent the local relationship) to measure the entirely asymmetric distribution of relationships between query images and classes. Deep Earth Mover's Distance (DeepEMD) [38] adopts the EMD metric to confirm image correlation by calculating the distance between image representations. Although DeepEMD [38] considers joint distribution, its computational cost is higher. The Feature Map Reconstruction Network (FRN) [39] uses ridge regression to directly rebuild query features from support features and uses the negative mean square Euclidean distance as a distance metric. Based on FRN [39], the foreground-aware FRN (FA-FRN) [2] uses the weighted squared Euclidean distance as a distance metric. The Bi-Similarity Network (BSNet) [40] uses two similarity metrics to simultaneously measure the similarity between images because a single similarity measure usually has restrictions in computing the similarity between images.

### B. Our Motivation

Based on the understanding of previous research, we conclude the following.

*1) In Terms of Feature Extraction:* Some methods have inherent flaws; for example, CAN [27] only considers spatial differences and omits channel differences, whereas PARN [28] only focuses on spatial location information. This observation implies that the features extracted using these methods are not sufficiently representative. Moreover, when only a relatively small amount of data is available for training, the previously mentioned methods commonly cause overfitting problems because of the multiple network parameters and complex structures, which can affect the final performance. Therefore, lightweight and effective feature extractors must be developed.

*2) In Terms of Metric Classification:* In contrast to the covariance-matrix-based methods commonly based on the covariance matrix as image representation, using the BDC matrix is a better option because this matrix does not consider the joint distribution, which can naturally handle nonlinear relationships between images and has the capability of statistical correlation modeling. This model also requires a lower computational cost than that of DeepEMD [38].

*3) In Terms of Baseline:* Based on the advantage of BDC matrix, Meta DeepBDC [37] has proved to be efficient and popular for target classification of natural optical images. This finding implies that Meta DeepBDC [37] may be a good baseline candidate for the further development of a potential target classification algorithm in optical remote sensing images as natural optical images and remote sensing optical ones have similar imaging mechanisms.

However, Meta DeepBDC [37] has some shortcomings in practical application:

1) First, it is insufficient to generate representative feature maps using only a backbone. Because the backbone interacts with all the regions of the image, the output features may comprise influence from irrelevant regions or interference from complex backgrounds.
2) Second, it only utilizes support BDC matrices to generate support prototypes and classifies the query images. However, we found that support prototype representations are typically not sufficiently discriminative to

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represent the differences in class distribution of the query images. This observation can be attributed to Meta DeepBDC [37], neglecting the abundant information about query images and the matching relationships between images.

Inspired by these inferences, with Meta DeepBDC [37] as the baseline, LATA module is added to the feature extractor and bi-prototype module is added to the metric classifier to resolve insufficient feature extraction ability and image discrimination ability, improving the fine-grained classification performance of optical images. The study focuses on the following.

1) In the feature extraction stage, to avoid overfitting caused by numerous network parameters, the LATA module does not introduce complex operators to represent the importance of the weights. However, LATA regards each feature map as a set of descriptors and utilizes appropriate energy functions to measure linear separability between the target and other descriptors, thereby generating a unique 3-D weight along the spatial and channel directions for each descriptor. The smaller the final derived energy value is, the more distinguished the target descriptors are from the surrounding descriptors. Thus, the network can automatically extract more representative features.
2) In the metric classification stage, the bi-prototype module makes full use of relevant information in the support and query images, and highly accurate prototypes are generated to improve the image discrimination ability. According to the matching relationship between the images, if the support prototypes can classify the query images, they can also classify the support images. Because the labels of the support images are known, the obtained support features can be positively transmitted to the support prototypes. We call this process positive transmission. Meanwhile, the bi-prototype module can also use the known query image information, i.e., the query prediction scores, as the query prototypes to reversely classify support images, because the number of query images is greater than that of support images. Furthermore, the query prediction scores contain both query and support features, which implies that the network can learn greater general features; thus, these features can be reversely transmitted to the support prototypes, further enhancing the accuracy of the prototypes. We call this process reverse transmission.

## III. Proposed Methodology

In this section, first, some elementary knowledge of few-shot classification is introduced; then, the LATA-BP-BDC is described as a whole. Moreover, the main modules of the LATA-BP-BDC are elaborated, respectively.

### A. Preliminary

Few-shot image classification is designed to learn how to distinguish C categories, but every category has only K-labeled samples, which is called the C-way K-shot task. With the purpose of learning new knowledge from a small number of samples, a dataset is grouped into the auxiliary set ($\mathcal{A}$) and evaluation set ($\mathcal{T}$), where $\mathcal{A} \cap \mathcal{T} = \phi$, which implies that $\mathcal{A}$ and $\mathcal{T}$ do not intersect. Sometimes, $\mathcal{A}$ is divided into training set ($\mathcal{A}_{\text{Train}}$) and validation set ($\mathcal{A}_{\text{Val}}$), where $\mathcal{A}_{\text{Train}} \cap \mathcal{A}_{\text{Val}} = \phi$. The model is trained on A and tested on $\mathcal{T}$. To avoid affecting the test accuracy, the label space of $\mathcal{A}$ cannot intersect with that of $\mathcal{T}$.

Currently, most few-shot classification methods utilize the episode training mechanism (ETM) [41]. Specifically, in a C-way K-shot task, the core of one episode is random selection of the samples in C categories to form a labeled support set $D^s = \{(x_i, y_i)\}_{i=1}^{CK}$ and an unlabeled query set $D^q = \{(q_i, y_i^q)\}_{i=1}^{CQ}$, where $D^s \cap D^q = \phi$, $x_i$ is the $i$th support image of $D^s$ and $y_i$ is the label, $q_i$ is the $i$th query image of $D^q$ and $y_i^q$ is the label, $y_i, y_i^q \in \{1, \ldots, C\}$, and $CK(CQ)$ is the number of support (query) images. $\mathcal{A}$ and $\mathcal{T}$ consist of a multitude of episodes. During the training stage, the model is trained on $\mathcal{A}$ to minimize the prediction error of the query set in every episode, and the generalization performance of the model is tested on $\mathcal{T}$ during the testing stage. The episodes used in training are the same as the settings in testing. The superiority of ETM is that it allows the model to learn, and constantly changing tasks allow the network to adapt to new few-shot tasks, making its generalization performance better and more suitable for FSL scenarios.

### B. Overall Architecture of the LATA-BP-BDC

The LATA-BP-BDC is shown in Fig. 2, which primarily consists of the following modules: backbone, LATA module, BDC module, metric module, and bi-prototype module. In particular, the LATA and bi-prototype modules are innovative modules. The first two modules form the feature extractor, and the last three modules form the metric classifier.

The process of the LATA-BP-BDC is mainly as follows.

First, the support images and the query images are fed into the feature extractor. The feature maps are generated through the backbone and sent to the LATA module to reduce the influence of irrelevant areas and background and enhance the features of key areas.

Next, the generated representative feature maps are sent to the metric classifier. First, BDC matrices are generated as image representation through BDC module; then, accurate and efficient class prototypes are generated by using bi-prototype module (including positive transmission and reverse transmission), which can better represent the features of classes.

Finally, in the metric module, the distance between the BDC matrix of the query image and various prototypes is calculated; then, the measurement results are fed into the softmax function to get the probability that the query image belongs to each category, thus completing the classification.

In addition, to better illustrate the whole process of the LATA-BP-BDC, Algorithm 1 is as follows.

### C. LATA Module

The essence of the attention mechanism is to utilize relevant feature maps to learn weight distribution and then apply the

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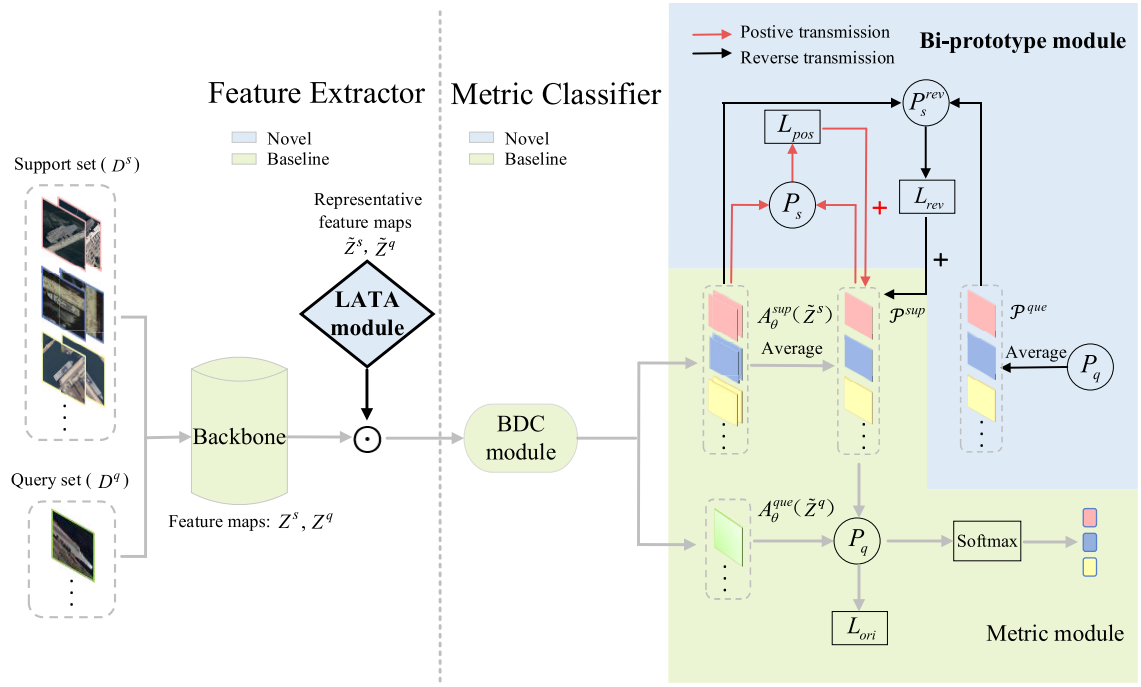



Fig. 2. Architecture overview of the LATA-BP-BDC. $A_\theta^{\text{sup}}(\tilde{Z}^s)$ and $A_\theta^{\text{que}}(\tilde{Z}^q)$ are support (query) BDC matrices. $\mathcal{P}^{\text{sup}}$ ($\mathcal{P}^{\text{que}}$) are support (query) prototypes. $P_s(P_s^{rev}\text{j}$ and $P_q$ are support and query predictions. In red line, we compute the distance between $A_\theta^{\text{sup}}$ and $\mathcal{P}^{\text{sup}}$ to get $P_s$. In black line, we generate $\mathcal{P}^{\text{que}}$ from $P_q$ to get $P_s^{rev}$. Two "+" indicate to enhance the accuracy of the prototypes.

**Algorithm 1** Model Training

**Input:** $\mathbb{T}$: distribution over tasks, $\xi$: learning rate, $\alpha(\beta)$ : trade-off hyperparameter
**Output:** model parameter $\eta$;

randomly initialize $\eta$;
**repeat**
  sample batch of tasks $t \sim \mathbb{T}$;
  **for** each task $t_i \in t$ **do**
    sample $\{D^s, D^q\} \sim t_i$;
    **Backbone:** calculate embeddings of $D^s$ and $D^q \rightarrow$ feature maps: $Z^s, Z^q$;
    **LATA:** get 3D weights using $Z^s, Z^q \rightarrow$ representative feature maps: $\tilde{Z}^s, \tilde{Z}^q$;
    **BDC:** obtain image representations$\rightarrow$ BDC matrices: $A_\theta^{\text{sup}}(\tilde{Z}^s)$, $A_\theta^{\text{que}}(\tilde{Z}^q)$;
    **Metric:** $A_\theta^{\text{sup}}(\tilde{Z}^s) \xrightarrow{avg}$ support prototypes $\mathcal{P}^{\text{sup}}$, calculate query predictions $P_q = dis(A_\theta^{\text{que}}(\tilde{Z}^q), P^{\text{sup}}) \rightarrow L_{\text{ori}}$;
    **Positive transmission:** calculate support predictions $P_s = dis(A_\theta^{\text{sup}}(\tilde{Z}^s), \mathcal{P}^{\text{sup}}) \rightarrow L_{\text{pos}}$;
    **Reverse transmission:** $P_q \xrightarrow{avg}$ query prototypes $\mathcal{P}^{\text{que}}$, reversely calculate support predictions $P_s^{rev} = dis(A_\theta^{\text{sup}}(\tilde{Z}^s), \mathcal{P}^{\text{que}}) \rightarrow L_{\text{rev}}$;
    $L = L_{\text{ori}} + \alpha L_{\text{pos}} + \beta L_{\text{rev}}$;
    Update model: $\eta \leftarrow \eta - \xi * \nabla_\eta L$;
  **end for**;
**until** convergence;

learned weights on top of the original feature maps for final weighting to compute more discriminative local features.

However, the methods of applying weights are slightly different: weighting can act on the channel (spatial) scale to weight different channel (spatial) features, thus generating 1-D (2)-D) weights, which treat equally descriptors of each channel (spatial position), limiting the network to learn more discriminative features [43]. Theoretically, 3-D weights (combining both channels and spatial dimensions) are superior to 1-D and 2-D weights.

Moreover, most existing attention mechanisms use common operators that represent the importance of weights, such as the convolutional block attention module (CBAM) [44], which adds the average pooling layer (GAP), maximum pooling layer (GMP), convolutional layer (Conv2D), fully connected layer (FC), and other operators to generate 1-D and 2-D weights, after which they are combined. Although a better performance is achieved, the number of parameters and computational complexity are increased. Unlike the existing channel and spatial attention modules, the LATA module generated 3-D weights for feature maps without adding the aforementioned operators to the original network.

Fig. 3 shows a schematic of the LATA module. Each feature map is regarded as a set of descriptors, and each descriptor describes the discriminant features of a specific spatial location. We introduce an energy function [43] to determine the target descriptor by measuring the linear separability between descriptors. The energy function of each descriptor is defined as follows:

$$E_n = \left(y_n - \hat{n}\right)^2 + \frac{\sum_{j=1}^{G-1}\left(y_0 - \hat{z}_j\right)^2}{G-1} \tag{1}$$

where $\hat{n} = w_n n + b_n$ and $\hat{z}_j = w_n z_j + b_n$ are the linear transforms of $n$ and $z_j$, where $n$ and $z_j$ denote the target

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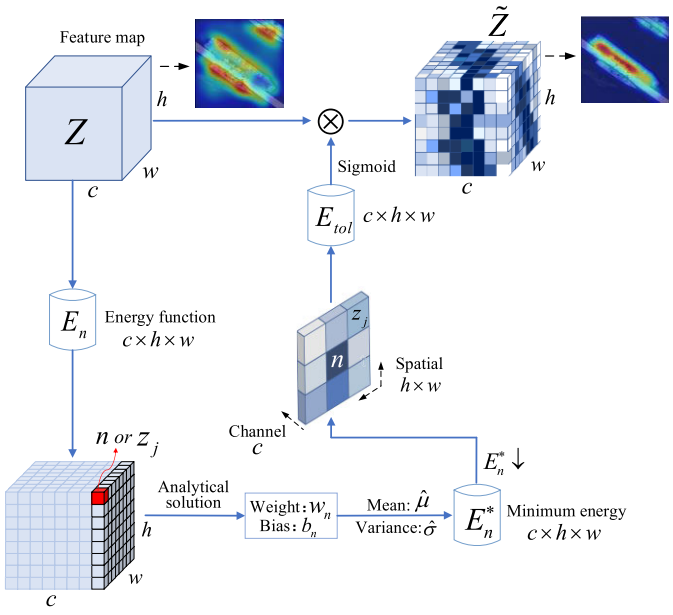



Fig. 3. Structure of LATA module. $n$ and $z_j$ denote the target descriptor and other descriptors in one channel of $Z$. Each $E_n$ assigns a unique 3-D weight to each descriptor. Smaller $E_n^*$, the more obvious the difference between $n$ and $z_j$. $E_{\text{tol}}$ groups all $E_n^*$ across channels and spatial dimensions. Color is darker, weight percentage is greater, and the importance is higher.

descriptor and other descriptors in one channel of $Z$. $Z \in R^{c\times h\times w}$ is the feature map obtained from the backbone, $c$ is the channel number of every descriptor, $G = h \times w$ is the number of descriptors on that channel, and each descriptor can correspond to a spatial position. The symbol $j$ is an index of the spatial dimension. $w_n$ is the weight, and $b_n$ is the bias. $\mathcal{Y}_n$ and $\mathcal{Y}_0$ are two different scalars.

Equation (1) is equivalent to determine the linear separability between $n$ and $z_j$ in the same channel by minimizing this equation. Specifically, (1) obtains the minimal value when $\hat{n}$ equals to $y_n$, and all $\hat{z}_j$ equal to $y_0$. For simplicity, (1) is simplified by using a binary notation (i.e., 1 and $-1$) for $y_n$ and $y_0$ and adding a regularizer. The final energy function is as follows:

$$E_n = (1-(w_n n+b_n))^2 + \frac{\sum_{j=1}^{G-1}\left(-1-\left(w_n z_j+b_n\right)\right)^2}{G-1} + \gamma w_n^2 \tag{2}$$

where $\gamma$ is a regularized hyperparameter.

Theoretically, each channel has $G$ energy functions, such that each descriptor can be assigned a unique weight. To render the model lightweight, (1) yields an analytical solution for $w_n$ and $b_n$, which is obtained by

$$w_n = -\frac{2(n-\mu_n)}{(n-\mu_n)^2+2\sigma_n^2+2\gamma}, b_n = -\frac{1}{2}(n+\mu_n)w_n \tag{3}$$

where $\mu_n = (1/G-1)\sum_{j=1}^{G-1} z_j$ and $\sigma_n^2 = (1/G-1)\sum_{j=1}^{G-1}(z_j-\mu_n)^2$ denote the mean and variance of all descriptors except $n$ in that channel, respectively. The solution of (2) is obtained for a single channel. In this case, it is assumed that all the descriptors in a particular channel follow the same distribution, which enables to calculate the mean and variance of all the descriptors and to reuse for all the descriptors in that channel. This assumption significantly reduces the computational cost and avoids the iterative calculation of $\mu$ and $\sigma$ at each spatial location. Therefore, the minimum energy can be computed as follows:

$$E_n^* = \frac{4\left(\hat{\sigma}^2+\gamma\right)}{\left(n-\hat{\mu}\right)^2+2\hat{\sigma}^2+2\gamma}. \tag{4}$$

where $\hat{\mu} = (1/G)\sum_{j=1}^{G} z_j$ and $\hat{\sigma}^2 = (1/G)\sum_{j=1}^{G}(z_j-\hat{\mu})^2$. According to (3), the smaller the value of $E_n^*$ is, the more evident the difference between $n$ and $z_j$ will be. Therefore, the importance of each descriptor is measured by $(1/E_n^*)$.

Subsequently, we performed feature refinement based on the definition of the attention mechanism. The optimization stage is as follows:

$$\tilde{Z} = \text{sigmoid}\left(\frac{1}{E_{\text{tol}}}\right) \otimes Z \tag{5}$$

where $E_{\text{tol}}$ groups all $E_n^*$ across channels and spatial dimensions. A sigmoid function is used to prevent the value of $E_{\text{tol}}$ from becoming enormous. $\otimes$ denotes the product sign.

Thus, the LATA module generates 3-D attention weights along the channel and spatial dimensions. Therefore, this model can adaptively extract more important feature maps $\tilde{Z} \in R^{c\times h\times w}$ and filter complex background interference by multiplying the original feature maps $Z$ by the generated 3-D attention weights.

### D. BDC Module

The BDC metric is proposed by Székely and Rizzo [45] and Székely et al. [46] and is interpreted as the Euclidean distance between the product of marginals and the joint characteristic function (JCF). The BDC metric is expressed as follows:

$$\rho(X,Y) = \int_{R^d}\int_{R^e} \frac{|\emptyset_{XY}(t,s)-\emptyset_X(t)\emptyset_Y(s)|^2}{r_d r_e \|t\|^{1+d}\|s\|^{1+e}} dtds \tag{6}$$

where $X \in R^d$ and $Y \in R^e$ denote the random vectors of dimension $d$ and $e$, respectively. $\emptyset_{XY}(t,s)$ is the JCF. $\|\cdot\|$ is the Euclidean norm, $r_d = (\pi^{(1+d/2)}/\Gamma((1+d/2)))$, and $\Gamma$ denotes the gamma function.

According to the theory of probability, $X$ and $Y$ are independent if and only if $\emptyset_{XY}(t,s) = \emptyset_X(t)\emptyset_Y(s)$. We assumed that the set of $f$ observations $\{(\mathbb{x}_1,\mathbb{y}_1),\ldots,(\mathbb{x}_f,\mathbb{y}_f)\}$ are independent and identically distributed. $\emptyset_{XY}(t,s) = (1/f)\sum_{k=1}^{f}\exp(\imath(t^T\mathbb{x}_k+s^T\mathbb{y}_k))$, $\imath$ is the imaginary unit, and $T$ indicates the matrix transpose.

Although (5) is complex, the BDC metric has a closed-form solution for discrete observations. Specifically, $\hat{A} = (\hat{a}_{kl}) \epsilon R^{f\times f}$ and $\hat{a}_{kl} = \|\mathbb{x}_k - \mathbb{x}_l\|$ is a Euclidean distance matrix calculated among the pairs of observations of $X$. Similarly, $\hat{B} = (\hat{b}_{kl}) \epsilon R^{f\times f} = \|\mathbb{y}_k - \mathbb{y}_l\|$. Subsequently, $A = (a_{kl})$ denotes the BDC matrix. $a_{kl} = \hat{a}_{kl} - (1/f)\sum_{k=1}^{f}\hat{a}_{kl} - (1/f)\sum_{l=1}^{f}\hat{a}_{kl} - (1/f^2)\sum_{k=1}^{f}\sum_{l=1}^{f}\hat{a}_{kl}$, where the second term is the mean of the $l$th column, the third term is the mean of the $k$th row, and last term is the mean of all elements of $\hat{A}$. $B$ can be calculated similarly. Moreover, $a$ and $b$ denote two BDC vectors, and because the BDC matrix is symmetric [46],

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it can be expressed as the inner product of $a$ and $b$. Therefore, the BDC metric can be written as follows:

$$\rho(X, Y) = tr\left(A^T B\right) = \langle a, b\rangle = a^T b \tag{7}$$

where $tr(\cdot)$ is the matrix trace. For $a(resp., b)$, it is acquired by the means of extracting the upper triangular part of $A(resp., B)$, followed by vectorization.

Meta DeepBDC [37] represents the BDC as a pooling layer that can be embedded in a network structure, which inputs the feature map and outputs the BDC matrix as the image representation. Therefore, the inner product of the two BDC matrices is the similarity between the two corresponding images.

Specifically, we assume that the network is parameterized by $\theta$. First, the representative feature maps $\tilde{Z}\epsilon R^{c\times h\times w}$ extracted from the LATA module are fed into the BDC module, where $c$, $h$, and $w$ represent the channel numbers, the spatial height, and the spatial width, respectively. Next, the tensor is reshaped to a matrix $\dot{Z}\epsilon R^{hw\times c}$, and an observation of random vector is viewed either from each column $\dot{z}_k\epsilon R^{hw}$ or each row (after transposition) $\dot{z}_g\epsilon R^{c}$. After analysis, using $\dot{z}_k$ as a random observation, three operators are used to compute the Euclidean distance square matrix $\bar{A} = (\bar{a}_{kl})$, where $\bar{a}_{kl}$ is the square of the Euclidean distance between the $k$th and $l$th columns of $\dot{Z}$, and $\hat{A} = (\bar{a}_{kl})^{1/2}$. The equations for $\bar{A}$ and $A$ are as follows:

$$\begin{aligned}\bar{A} &= 2\left(\mathbf{1}\left(\dot{Z}^T\dot{Z}\circ \boldsymbol{I}\right)\right)_{\text{sym}} - 2\dot{Z}^T\dot{Z}\\ A &= \hat{A} - \frac{2}{c}\left(\mathbf{1}\hat{A}\right)_{\text{sym}} + \frac{1}{c^2}\mathbf{1}\hat{A}\mathbf{1}\end{aligned} \tag{8}$$

where $\mathbf{1}\in R^{c\times c}$ is a matrix and each element is 1, $\circ$ represents the Hadamard product, $\boldsymbol{I}$ denotes the identity matrix, and $(U)_{\text{sym}} = (1/2)(U + U^T)$. Next, $A_\theta(\tilde{Z})$ indicates the BDC matrix, which is calculated by the $\theta$-parameterized network with the input feature maps $\tilde{Z}$.

### *E. Bi-Prototype Module*

The metric module computes the distances to every class prototype to perform the classification. In a task $(D^s, D^q)$, the support image $x_i \in D^s$ is fed to the network to generate the support BDC matrix $A_\theta^{\text{sup}}(\tilde{x}_i)$, and the query image $q_i \in D^q$ is fed to the network to generate the query BDC matrix $A_\theta^{\text{que}}(\tilde{q}_i)$, where $\tilde{x}_i(\tilde{q}_i)$ denotes a support (query) image with representative features. Subsequently, the support prototype of $m$th class ($\mathcal{P}_m^{\text{sup}}$) denotes the mean of $A_\theta^{\text{sup}}(\tilde{x}_i)$ that belongs to the class

$$\mathcal{P}_m^{\text{sup}} = \frac{1}{K}\sum_{\tilde{x}_i\in S_m} A_\theta^{\text{sup}}\left(\tilde{x}_i\right) \tag{9}$$

where $S_m$ denotes the sample set marked as the $m$th class in $D^s$. We calculate the distance between $A_\theta^{\text{que}}(\tilde{q}_i)$ and $\mathcal{P}_m^{\text{sup}}$ and forecast the query images by using a softmax function. The score for predicting query image $q_i$ as the $m$th class is as follows:

$$P_q(y = m \mid q_i) = \frac{\exp\left(\tau tr\left(A_\theta^{\text{que}}\left(\tilde{q}_i\right)^T\mathcal{P}_m^{\text{sup}}\right)\right)}{\sum_{m=1}^{C}\exp\left(\tau tr\left(A_\theta^{\text{que}}\left(\tilde{q}_i\right)^T\mathcal{P}_m^{\text{sup}}\right)\right)} \tag{10}$$

where $\tau$ is the learnable proportion parameter.

Consequently, the original loss function for each task can be expressed as follows:

$$L_{\text{ori}} = -\frac{1}{CQ}\sum_{i=1}^{CQ}\log\left(P_q(y = m|q_i)\right). \tag{11}$$

However, the aforementioned support prototypes ($\mathcal{P}^{\text{sup}}$) are usually insufficiently discriminative to represent the differences in the class distribution of queries. Therefore, based on the insights obtained from [47], we use a bi-prototype module to generate highly accurate prototypes. The bi-prototype module is divided into two stages: a positive transmission stage and a reverse transmission stage.

In the positive transmission stage, in the same way as the predicted query images, we use the distance between $A_\theta^{\text{sup}}(\tilde{x}_i)$ and $\mathcal{P}^{\text{sup}}$ to obtain the predicted score of the support images. The predicted score of support image $x_i$ is as follows:

$$P_s(y = m \mid x_i) = \frac{\exp\left(\tau tr\left(A_\theta^{\text{sup}}\left(\tilde{x}_i\right)^T\mathcal{P}_m^{\text{sup}}\right)\right)}{\sum_{m=1}^{C}\exp\left(\tau tr\left(A_\theta^{\text{sup}}\left(\tilde{x}_i\right)^T\mathcal{P}_m^{\text{sup}}\right)\right)}. \tag{12}$$

The predicted loss can then be computed as the mean of the negative logarithmic probabilities of all the support images in each task. Because the labels of the support images are known, the generated $L_{\text{pos}}$ can lead the network to positively transmitted to the known information to $\mathcal{P}^{\text{sup}}$ to achieve representative prototypes. $L_{\text{pos}}$ is formulated as follows:

$$L_{\text{pos}} = -\frac{1}{CK}\sum_{i=1}^{CK}\log(P_s(y = m|x_i)). \tag{13}$$

In the reverse transmission stage, to entirely use the known information, we attempt mining the information in $P_q$. Specifically, $P_q$ of each class is regarded as the query prototype ($\mathcal{P}^{\text{que}}$) for that class, and the distance between $A_\theta^{\text{sup}}(\tilde{x}_i)$ and $P^{\text{que}}$ is calculated to predict the support images in reverse. The predicted score is expressed as follows:

$$P_s^{rev}(y = m \mid x_i) = \frac{\exp\left(\tau tr\left(A_\theta^{\text{sup}}\left(\tilde{x}_i\right)^T\mathcal{P}_m^{\text{que}}\right)\right)}{\sum_{m=1}^{C}\exp\left(\tau tr\left(A_\theta^{\text{sup}}\left(\tilde{x}_i\right)^T\mathcal{P}_m^{\text{que}}\right)\right)} \tag{14}$$

where $\mathcal{P}_m^{\text{que}} = (1/CQ^m)\sum_{i=1}^{CQm} P_q(y = m \mid q_i^m)$ denotes the query prototype of the $m$th class, which is computed to predict the query image as the average of all probabilities of the $m$th class. $CQ^m$ indicates the query image numbers of the $m$th class in $D^q$ and $q_i^m$ denotes the query image of the $m$th class.

Thus, $\mathcal{P}^{\text{que}}$ contains valuable information about both the query and support images, and the prototype is modified in reverse by affecting the network's prediction of the query images. Therefore, the generated $L_{\text{rev}}$ forces the network to acquire more generic features in reverse, thereby facilitating the generation of accurate prototypes. The corresponding predicted loss is as follows:

$$L_{\text{rev}} = -\frac{1}{CK}\sum_{i=1}^{CK}\log\left(P_s^{rev}(y = m|x_i)\right). \tag{15}$$

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Finally, the whole model updates the gradient using a linear combination of these three losses

$$L = L_{\text{ori}} + \alpha L_{\text{pos}} + \beta L_{\text{rev}} \tag{16}$$

where $\alpha$ and $\beta$ are the tradeoff hyperparameters. The network can be trained end-to-end by optimizing $L$ using a gradient-descent algorithm. Moreover, because $L_{\text{pos}}$ contains information only on the support images, $L_{\text{rev}}$ also contains information on the support and query images. Therefore, the auxiliary constraint strength of $L_{\text{rev}}$ is larger and the weight proportion should be greater.

In other words, because the bi-prototype module can generate highly accurate prototypes, further calibrating the information differences between images can effectively enhance the correlation of the same category and the separability of different categories, thus forcing the network to achieve preferable classification performance.

## IV. Experiments

### A. Experimental Settings

*1) Dataset:* We utilize a public fine-grained ship dataset, the FGSCR dataset [48], to validate the availability of our proposed approach. The dataset is composed of 9320 optical remote sensing images, with 42 categories in total, and the size of the images ranges from approximately 50 × 50 to approximately 1500 × 1500. Different subclasses of the same parent class in the dataset may have different aspect ratios, resulting in large intraclass and small interclass discrepancies in the ship images. Moreover, in several practical scenarios, different types of ships may dock together; therefore, multiple ships are present in an image that interfere with ship recognition. Compared to other FGVC datasets, the FGSCR dataset is more appropriate for fine-grained classification because it suitably depicts a real scene. To ensure fairness and validity of the experimental results, we choose the same classification mode as that in [2]. As shown in Table I, we use 25 classes, including 15/5/5, for training/validation/evaluation. Fig. 4 shows the optical images of each ship type.

Although the present study focused on ship fine-grained classification of optical remote sensing images, however, one possible concern is whether the LATA-BP-BDC is a general method for solving other few-shot fine-grained image (such as natural images) classification problems. In other words, does the LATA-BP-BDC have good generalizability? Therefore, to solve this concern, it is necessary to validate the effectiveness of the LATA-BP-BDC on a general dataset for fine-grained few-shot classification. In this study, we select one of the most frequently used datasets in the FSL-FGVC field, the CUB dataset [49], similar to that in [2]. The CUB dataset contains 200 classes and 11 788 images in total. The image size varies from approximately 200 × 200 to approximately 500 × 500. We randomly divide the categories into 100 training, 50 validation, and 50 evaluation categories.

*2) Backbone Networks:* We select two frequently used backbone networks: Conv4 and ResNet12. Following the prior pattern, the size of the input images for Conv4 is 84 × 84. Conv4 is composed of four blocks, and each block consists of a 3 × 3 convolution with 64 channels, a batch normalization (BN), and a linear rectification function (ReLU); the first three blocks downsample the feature maps using 2 × 2 max-pooling. The size of the input images for ResNet12 is 84 × 84. ResNet12 comprises four stages, each containing one residual block, which has four stages with widths of [64, 160, 320, 640] and three 3 × 3 convolutional layers, a BN, and a Leaky-ReLU of 0.1. Similarly, the first three blocks downsample the feature maps by utilizing 2 × 2 max-pooling. To further validate the availability of our model, we also use the backbone network ResNet18 with an image input size of 224 × 224. First, ResNet18 uses a 7 × 7 convolution, BN, ReLU, and 2 × 2 max-pooling. ResNet18 comprises four stages, each containing one residual block, which has four stages with widths of [64, 128, 256, 512]. Similar to [34], [36], and [37], to acquire more convolutional features, we eliminate the last layer of downsampling.

TABLE I
Distribution of the FGSCR Dataset

| Data set | Ship class | Number |
|---|---|---|
| Training set | Ticonderoga | 608 |
| | Arleigh | 581 |
| | Nimitz | 554 |
| | Container | 455 |
| | Wasp | 453 |
| | Medical | 322 |
| | Independence | 210 |
| | Midway | 208 |
| | Freedom | 177 |
| | Tank | 160 |
| | Osumi | 116 |
| | Abukuma | 79 |
| | KittyHawk | 69 |
| | Sacramento | 51 |
| | Hyuga | 24 |
| Validation set | Whitby | 279 |
| | Sand | 260 |
| | Megayacht | 187 |
| | Kidd | 68 |
| | Murasame | 64 |
| Evaluation set | Antonio | 320 |
| | Crane | 143 |
| | Asagiri | 71 |
| | Kuznetsov | 68 |
| | Lzumo | 63 |

*3) Experimental Setup:* During training, we use standard data enhancement for all benchmarks, including random resized crops, random horizontal flips, and color jittering. The stochastic gradient descent (SGD) algorithm is used, which has a 0.9 momentum and a $5 \times 10^{-4}$ weight decay to train our proposed network. We uniformly sample the tasks from the training set to train the network. The initialization weights are obtained by pretraining the network on the training set. We randomly choose five classes, each with one support image (five support images) and 16 query images in a five-way one-shot (five-shot) assignment. During the evaluation process, we uniformly sample tasks from the evaluation set to assess the network performance. We adopt 15 query images in a five-way one-shot/five-shot setting. The regularized hyperparameter

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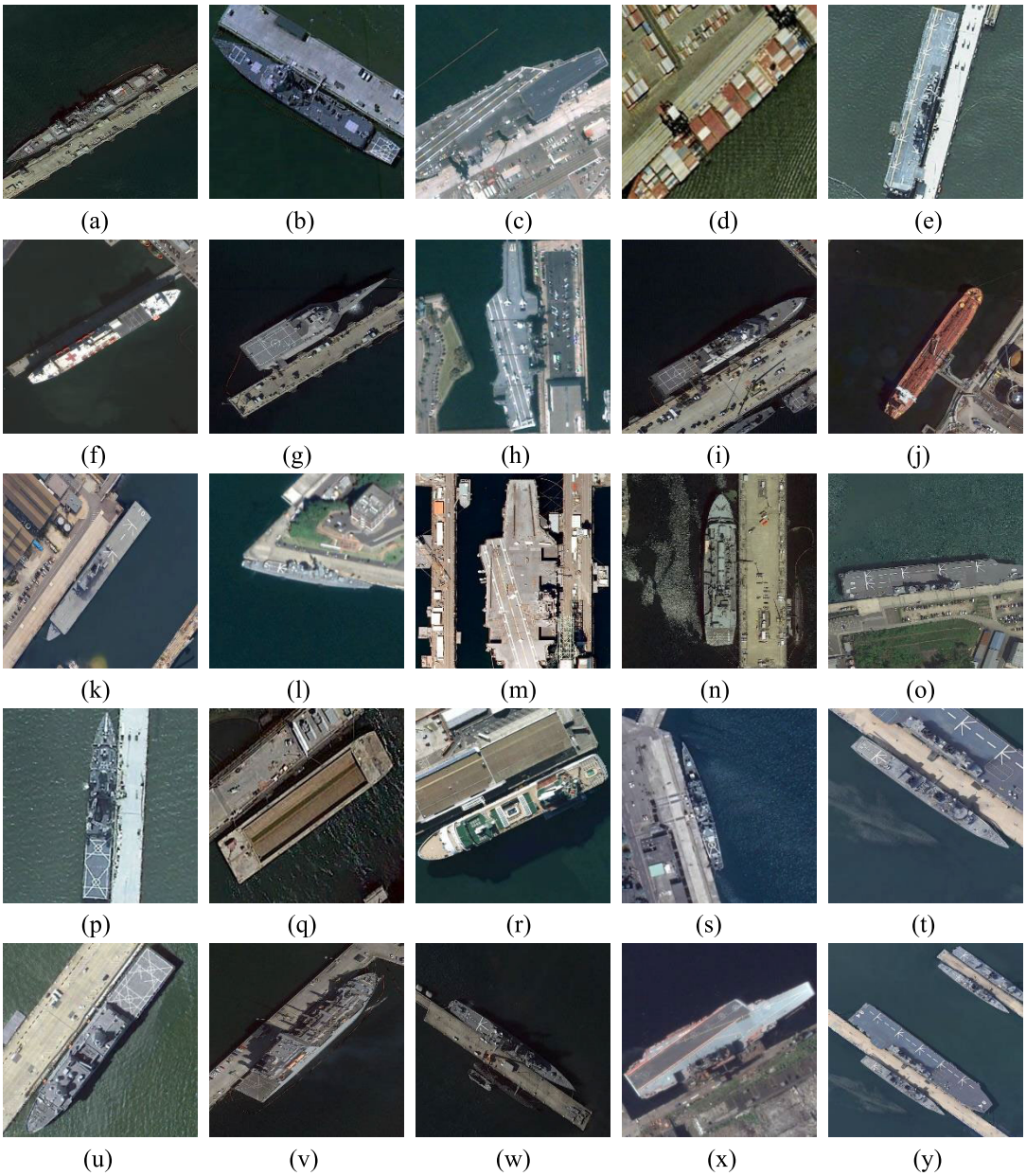



Fig. 4. Twenty-five ship types in the FGSCR dataset. (a) Ticonderoga. (b) Arleigh. (c) Nimitz. (d) Container. (e) Wasp. (f) Medical. (g) Independence. (h) Midway. (i) Freedom. (j) Tank. (k) Osumi. (l) Abukuma. (m) KittyHawk. (n) Sacramento. (o) Hyuga. (p) Whitby. (q) Sand. (r) Megayacht. (s) Kidd. (t) Murasame. (u) Antonio. (v) Crane. (w) Asagiri. (x) Kuznetsov. (y) Lzumo.

[$\gamma$ in (2)] is set to $1 \times 10^{-4}$, and the tradeoff hyperparameters [$\alpha$ and $\beta$ in (15)] are set to $\alpha = 1$ and $\beta = 3$ ($\alpha = 3$ and $\beta = 7$) in five-way one-shot/five-shot setting. All hyperparameters are cross-validated in the validation sets and fixed in all the experiments. We record the mean accuracy and 95% confidence intervals for 2000 episodes. All the experiments utilize the PyTorch framework and NVIDIA GeForce RTX3090Ti processor.

### B. Comparison With Advanced Methods

Because the LATA-BP-BDC belongs to the metric-based few-shot methods, we compare it to other metric-based methods, including the popular methods: ProtoNet [33], RelationNet [34], FRN [39], ReNet [31], and MATANet [32]; the latest FA-FRN [2]; and our baseline Meta DeepBDC [37].

*1) Comparison of Classification Performance Using FGSCR Dataset:* Table II reports the five-way classification accuracies of the FGSCR dataset using different methods for the backbone networks Conv4 and ResNet12. We reproduce the model in [33] with our training mode and network backbone, whereas we use publicly available codes for the methods in [32], [34], and [39] to obtain results and report the highest accuracies among our results and the published results. When Conv4 is used as the backbone, LATA-BP-BDC achieves the best performance. Compared with our baseline Meta DeepBDC [37], LATA-BP-BDC achieves a performance increase of 5.38% (one-shot) and 5.64% (five-shot). For the most advanced method, FA-FRN [2], LATA-BP-BDC achieves performance improvements of 4.61% (one-shot) and 0.91% (five-shot). When ResNet12

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is used as the backbone, LATA-BP-BDC gains the best performance. The accuracies of LATA-BP-BDC are 3.46% (one-shot) and 2.11% (five-shot) higher than those of Meta DeepBDC [37]. Moreover, the accuracies of LATA-BP-BDC are 1.7% (one-shot) and 0.83% (five-shot) higher than that of FA-FRN [2]. We also use ResNet18 as the backbone to obtain a performance boost. Compared with Meta DeepBDC [37], LATA-BP-BDC obtains 2.83% (one-shot) and 3.08% (five-shot) performance enhancements. However, for ProtoNet [33], the performance decreases as the number of backbone network layers increases. Therefore, for small datasets, such as FGSCR, LATA-BP-BDC achieves the most excellent performance not only in shallow networks but also in relatively deep network layers, indicating that LATA-BP-BDC is simple, easily available, and has a powerful classification capability.

*2) Comparison of Classification Performance Using CUB Dataset:* Table III reports the five-way classification accuracies of the CUB dataset using different approaches for the backbone networks Conv4 and ResNet12. We reproduced the model in [33] using our training mode and network backbone. Because a significant amount of research has been conducted on the CUB dataset [2], we directly adopt the published results of the dataset. When Conv4 is used as the backbone, LATA-BP-BDC achieves the best performance. Compared with that in Meta DeepBDC [37], LATA-BP-BDC acquires a performance improvement of 2.96% (one-shot) and 3.27% (five-shot). When ResNet12 is used as the backbone, LATA-BP-BDC gains the best performance. Compared with that in Meta DeepBDC [37], LATA-BP-BDC obtains a 2.09% (one-shot) and 1.71% (five-shot) performance gain. We also utilize ResNet18 as the backbone, and LATA-BP-BDC achieves the best performance. The accuracies of LATA-BP-BDC are 1.38% (one-shot) and 1.12% (five-shot) higher than those of Meta DeepBDC [37]. Thus, the LATA-BP-BDC acquires an outstanding classification ability not only for fine-grained optical remote sensing images but also for natural optical images, which validates the availability of the LATA-BP-BDC.

### C. Ablation Study

We perform ablation experiments to verify the feasibility of the major part of the LATA-BP-BDC. Meta DeepBDC [37] is the baseline method used for comparison.

*1) LATA Module:* We use ResNet12 and perform experiments on the FGSCR and CUB datasets to demonstrate the effectiveness of the LATA module, as well as a complexity comparison. Table IV shows the experimental results with the LATA module (Y) and without the LATA module (N). Because the baseline Meta DeepBDC [37] uses the inner product distance (one-shot) and Euclidean distance (five-shot) as distance functions, the same distance function is used here for fairness. For the FGSCR dataset, the addition of the LATA module to the network improves the performance by 1.7% (one-shot) and 1.55% (five-shot) compared with that when it is not added. For the CUB dataset, the performance improves by 1.92% (one-shot) and 0.99% (five-shot). Therefore, the LATA module achieves good performance on the FGSCR small dataset and the CUB large dataset.

To verify that LATA module is lightweight, we compared it with some potential modules, such as CBAM [44], Shuffle attention module (SANet) [50], and normalization-based module (NAM) [51]. Similar to the LATA module, these modules combine the spatial and channel attentions. However, the difference is that these modules express the importance of weights by building complex operators. Our module does not introduce the corresponding operators, but it generates 3-D weights by introducing the energy function. Therefore, to demonstrate the cost of the attention module, we report the parameter number (pn), floating-point operation number (GFLOPs), accuracy, and mean CPU inference time (CIT) for five-way one-shot/five-shot settings with 16 query images for each class.

As shown in Table V, CBAM [44] adds operators, such as GAP, GMP, Conv2D, and FC, to generate 1-D and 2-D weights from the channel and spatial levels, respectively. Although a good performance has been achieved, it has increased the pn and GFLOPs, and the running time is the longest. SANet [50] introduces feature grouping and channel permutation to reduce computational complexity; however, it still uses GAP to embed global information and group normalization (GN) to obtain spatial mode statistics. Although NAM [51] avoids adding complex FC and Conv2D, it uses the proportional factor of the BN to represent the importance of the weights. As the number of network layers increases, these modules increase the pn and GFLOPs.

Because our module does not introduce operators, such as those mentioned above, it does not increase the pn and GFLOPs, making it a lightweight attention module. Our module generates 3-D weights by introducing an energy function to calculate the linear separability between the target and other descriptors. The smaller the resulting energy value is, the more distinct the target features are from the surrounding features. Therefore, the model could automatically extract more representative features.

Notably, by comparing the CIT, although the computational cost of each module in all the methods increases, our module has the lowest computational cost. Our module primarily calculates the mean ($\mu$) and variance ($\sigma$) to obtain the 3-D weights, which results in corresponding computational losses. However, we have an analytical solution for 3-D weights that does not require repeated calculation of $\mu$ and $\sigma$ in each space. This solution significantly reduces computational costs, thereby effectively reducing computational losses. Moreover, the module achieves excellent classification results. Therefore, the proposed module is lightweight and efficient.

*2) Distance Function:* As the effectiveness of the attention module is verified above, subsequent experiments are conducted using the attention module. We use ResNet12 as the backbone. Currently, the most commonly used distance functions are the Euclidean, cosine, Manhattan, Chebyshev, and inner product distances, as mentioned in the baseline approach. Among these, although the baseline Meta DeepBDC [37] experimentally verifies that the performance of the Euclidean distance is higher than that of the cosine and inner product distances in the five-shot setting, the performance of the inner product and cosine distances is much higher than that of

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TABLE II
COMPARISON WITH ADVANCED METHODS ON THE FGSCR DATASETS. †REPRODUCED WITH OUR SETTING

| Method | Venue | Backbone | 5-way 1-shot | 5-way 5-shot |
|---|---|---|---|---|
| ProtoNet [33]† | NeurIPS'17 | Conv4 | 40.10±0.34 | 62.16±0.32 |
| RelationNet [34] | CVPR'18 | Conv4 | 42.83±0.16 | 56.24±0.12 |
| FRN [39] | CVPR'21 | Conv4 | 45.47±0.17 | 71.20±0.13 |
| MATANet [32] | ICPR'22 | Conv4 | 44.70±0.60 | 74.50±0.43 |
| FA-FRN [2] | TGRS'22 | Conv4 | 46.76±0.17 | 74.58±0.12 |
| Meta DeepBDC [37] | CVPR'22 | Conv4 | 45.99±0.36 | 69.85±0.32 |
| **Ours** | Ours | Conv4 | **51.37±0.39** | **75.49±0.26** |
| ProtoNet [33]† | NeurIPS'17 | Resnet12 | 49.19±0.38 | 71.11±0.28 |
| ReNet [31] | ICCV'21 | Resnet12 | 41.93±0.39 | 62.97±0.30 |
| FRN [39] | CVPR'21 | Resnet12 | 54.49±0.17 | 76.80±0.13 |
| FA-FRN [2] | TGRS'22 | Resnet12 | 56.91±0.16 | 79.50±0.11 |
| Meta DeepBDC [37] | CVPR'22 | Resnet12 | 55.15±0.35 | 78.22±0.29 |
| **Ours** | Ours | Resnet12 | **58.61±0.37** | **80.33±0.26** |
| ProtoNet [33]† | NeurIPS'17 | Resnet18 | 47.23±0.39 | 68.62±0.29 |
| Meta DeepBDC [37] | CVPR'22 | Resnet18 | 56.58±0.36 | 78.37±0.27 |
| **Ours** | Ours | Resnet18 | **59.41±0.38** | **81.45±0.26** |

TABLE III
COMPARISON WITH ADVANCED METHODS ON THE CUB DATASETS. †REPRODUCED WITH OUR SETTING

| Method | Venue | Backbone | 5-way 1-shot | 5-way 5-shot |
|---|---|---|---|---|
| ProtoNet [33]† | NeurIPS'17 | Conv4 | 62.42±0.51 | 79.80±0.33 |
| RelationNet [34] | CVPR'18 | Conv4 | 62.45±0.98 | 76.11±0.69 |
| CovaMNet [35] | AAAI'19 | Conv4 | 60.58±0.69 | 74.24±0.68 |
| FRN [39] | CVPR'21 | Conv4 | 63.84±0.22 | 78.13±0.17 |
| MATANet [32] | ICPR'22 | Conv4 | 65.84±0.69 | 83.78±0.45 |
| FA-FRN [2] | TGRS'22 | Conv4 | 65.91±0.22 | 80.87±0.16 |
| Meta DeepBDC [37] | CVPR'22 | Conv4 | 69.15±0.48 | 84.76±0.29 |
| **Ours** | Ours | Conv4 | **72.11±0.47** | **88.03±0.25** |
| ProtoNet [33]† | NeurIPS'17 | Resnet12 | 79.31±0.44 | 89.02±0.23 |
| RelationNet [34] | CVPR'18 | Resnet12 | 70.20±0.84 | 84.28±0.46 |
| DeepEMD [38] | CVPR'20 | Resnet12 | 75.65±0.83 | 88.69±0.50 |
| ReNet [31] | ICCV'21 | Resnet12 | 79.49±0.44 | 91.11±0.24 |
| FRN [39] | CVPR'21 | Resnet12 | 78.07±0.21 | 89.33±0.12 |
| FA-FRN [2] | TGRS'22 | Resnet12 | 78.72±0.20 | 91.33±0.11 |
| Meta DeepBDC [37] | CVPR'22 | Resnet12 | 80.27±0.43 | 91.71±0.21 |
| **Ours** | Ours | Resnet12 | **82.36±0.41** | **93.42±0.18** |
| ProtoNet [33]† | NeurIPS'17 | Resnet18 | 79.85±0.44 | 90.01±0.23 |
| RelationNet [34] | CVPR'18 | Resnet18 | 68.58±0.94 | 84.05±0.56 |
| ADM [36] | IJCAI'20 | Resnet18 | 79.31±0.43 | 90.69±0.21 |
| Meta DeepBDC [37] | CVPR'22 | Resnet18 | 81.23±0.42 | 92.43±0.19 |
| **Ours** | Ours | Resnet18 | **82.61±0.41** | **93.55±0.18** |

TABLE IV
ABLATION STUDY OF ATTENTION MODULE ON THE FGSCR AND CUB DATASET

| Dataset | Attention | 1-shot | 5-shot |
|---|---|---|---|
| FGSCR | N | 55.15±0.35 | 78.22±0.29 |
| | Y | **56.85±0.37** | **79.77±0.28** |
| CUB | N | 80.27±0.43 | 91.71±0.21 |
| | Y | **82.19±0.42** | **92.70±0.19** |

TABLE V
COMPLEXITY COMPARISONS

| Attention | PN (M) | GFLOPs | CIT | | Accuracy | |
|---|---|---|---|---|---|---|
| | | | 1-shot | 5-shot | 1-shot | 5-shot |
| Backbone | 12.41 | 3.51 | 1.74s | 2.14s | 55.15±0.35 | 78.22±0.29 |
| CBAM | 12.48 | 3.55 | 2.53s | 3.11s | 55.18±0.38 | 79.26±0.28 |
| SANet | 12.41 | 3.51 | 2.08s | 2.47s | 55.78±0.40 | 78.73±0.29 |
| NAM | 12.41 | 3.51 | 2.21s | 2.72s | 56.79±0.42 | 78.65±0.28 |
| Our | 12.41 | 3.51 | 2.01s | 2.42s | **56.85±0.37** | **79.77±0.28** |

the Euclidean distance in the one-shot setting. Thus, result indicates that the Euclidean distance is not sufficiently robust and fluctuates significantly.

However, Oreshkin et al. [42] demonstrated that the interaction between the distance function and softmax can be scaled to increase the performance of the distance function. Therefore, we use a scaled cosine (cosine*) distance to perform the experiments. Fig. 5 shows the accuracy of these distance functions with five-way five-shot setting in the pretraining phase on the FGSCR dataset. It is clearly seen that the cosine* distance

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TABLE VI
ABLATION STUDY OF SEVERAL COMMON DISTANCE METRICS ON THE FGSCR AND CUB DATASET

| Dataset | Metric | 1-shot | 5-shot |
|---|---|---|---|
| FGSCR | Euclidean | 56.03±0.38 | 79.77±0.28 |
| | Inner | 56.85±0.37 | 77.22±0.28 |
| | Cosine* | **57.46±0.38** | **80.11±0.26** |
| CUB | Euclidean | 77.09±0.45 | 92.70±0.19 |
| | Inner | 82.19±0.42 | 91.99±0.22 |
| | Cosine* | **82.32±0.40** | **93.44±0.18** |

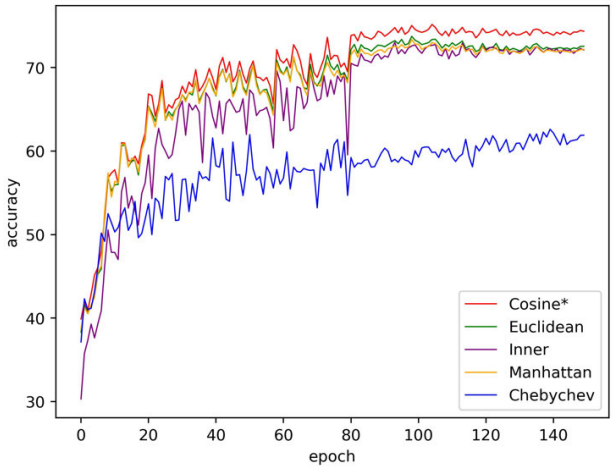



Fig. 5. Ablation study of different distance functions with five-way five-shot accuracy on the FGSCR dataset.

is higher than that of the remaining four commonly used distance methods. Table VI presents the experimental results of the three distance functions for the FGSCR and CUB datasets. We further verified that the Meta DeepBDC [37] has a higher inner product distance performance than that of the Euclidean distance in the one-shot setting and a higher Euclidean distance performance than that of the inner product distance in the five-shot setting in the experiment. However, the performance of the cosine* distance is higher than that of the inner product distance and Euclidean distance in both one-shot and five-shot tasks. For the FGSCR and CUB datasets, the cosine* distance outperforms the Euclidean distance by 1.43% and 5.23% for the one-shot setting, respectively, and the cosine* distance outperforms the inner product distance by 2.89% and 1.45% for the five-shot setting, respectively. Therefore, the cosine* distance is a more suitable and effective choice for the FGSCR and CUB datasets.

*3) Bi-Prototype Module:* The cosine* distance is used for all distance functions to ensure fairness. As shown in Table VII, when the backbone network on the FGSCR dataset is ResNet12, for the one-shot setting, the bi-prototype module outperforms the original prototype module by 1.15%; for the five-shot setting, a slight improvement is achieved. When the backbone network is ResNet18, the bi-prototype module outperforms the original prototype module by 0.63% (one-shot) and 0.55% (five-shot). This finding is consistent with previous analyses because the dataset is not uniformly distributed per class and has a relatively small amount of data,

TABLE VII
ABLATION STUDY OF BI-PROTO MODULE ON THE FGSCR DATASET

| Backbone | Method | 1-shot | 5-shot |
|---|---|---|---|
| Resnet12 | Proto | 57.46±0.38 | 80.11±0.26 |
| | Bi-Proto | **58.61±0.37** | **80.33±0.26** |
| Resnet18 | Proto | 58.78±0.36 | 80.90±0.26 |
| | Bi-Proto | **59.41±0.38** | **81.45±0.26** |

TABLE VIII
ABLATION STUDY OF BI-PROTO MODULE ON THE CUB DATASET

| Backbone | Method | 1-shot | 5-shot |
|---|---|---|---|
| Resnet12 | Proto | 82.32±0.40 | **93.44±0.18** |
| | Bi-Proto | **82.36±0.41** | 93.42±0.18 |
| Resnet18 | Proto | 82.42±0.40 | **93.58±0.18** |
| | Bi-Proto | **82.61±0.41** | 93.55±0.18 |

allowing the bi-prototype module to perform better and capture extremely accurate image information. In addition, as shown in Table VIII, a slight improvement of classification accuracy can be obtained for the CUB dataset. This result could be attributed to the fact that the CUB dataset is evenly distributed and contains a large amount of data, and the original prototype module could obtain sufficient image information. We concluded that the bi-prototype module is more suitable for small datasets with an uneven distribution, but it is also competitive for large datasets with a uniform distribution.

### D. Visualization Results

*1) Visualization of LATA Module:* To provide insights into the influence of the LATA module on the performance of LATA-BP-BDC, some samples of the evaluation set visualization are provided. The visualization samples are implemented using Grad-CAM [52]. Darker regions in the feature maps indicate that the corresponding regions are more categorically significant. As shown in Fig. 6, the model with the LATA module can suitably extract discriminative information from ship images. We can carefully observe the following.

1) When the target region occupied a larger area, as in Kuznetsov (first column), Antonio (fourth column), and Crane (fifth column), the model with the LATA module identifies a larger area and darker color of the target region than that without it.
2) When the share of the background region is relatively large, as in Asagiri (second column), the model without the LATA attention module recognizes all the background regions, whereas the model with the LATA module could accurately identify the target region.
3) When multiple ships are present in an image, as in Lzumo (third column), the model without the LATA module identifies all ships, principally the nontarget ones. However, the model with the LATA module is capable of accurately identifying the target ship region.

Based on the above analysis, the model with the LATA module can accurately capture the target regions in each image, which

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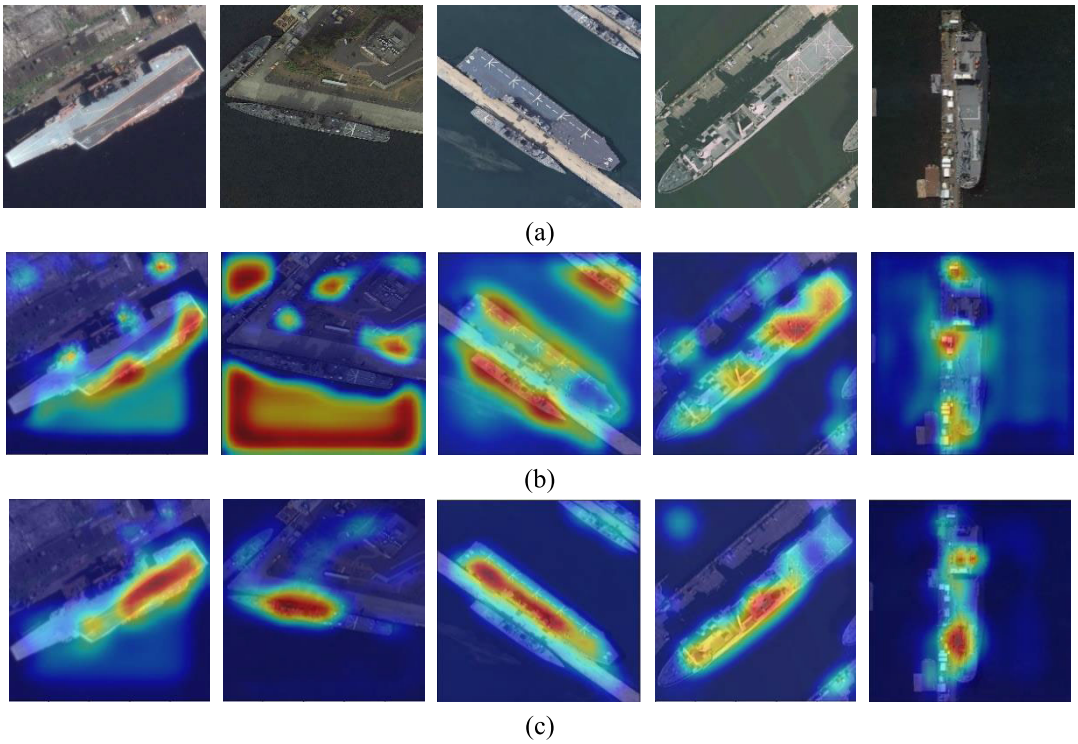


Fig. 6. Original ship images and the attention maps from GradCAM. (a) Original ship images: Kuznetsov, Asagiri, Lzumo, Antonio, and Crane (from left to right). (b) Grad-cam diagram without LATA module. (c) Grad-cam diagram with LATA module.

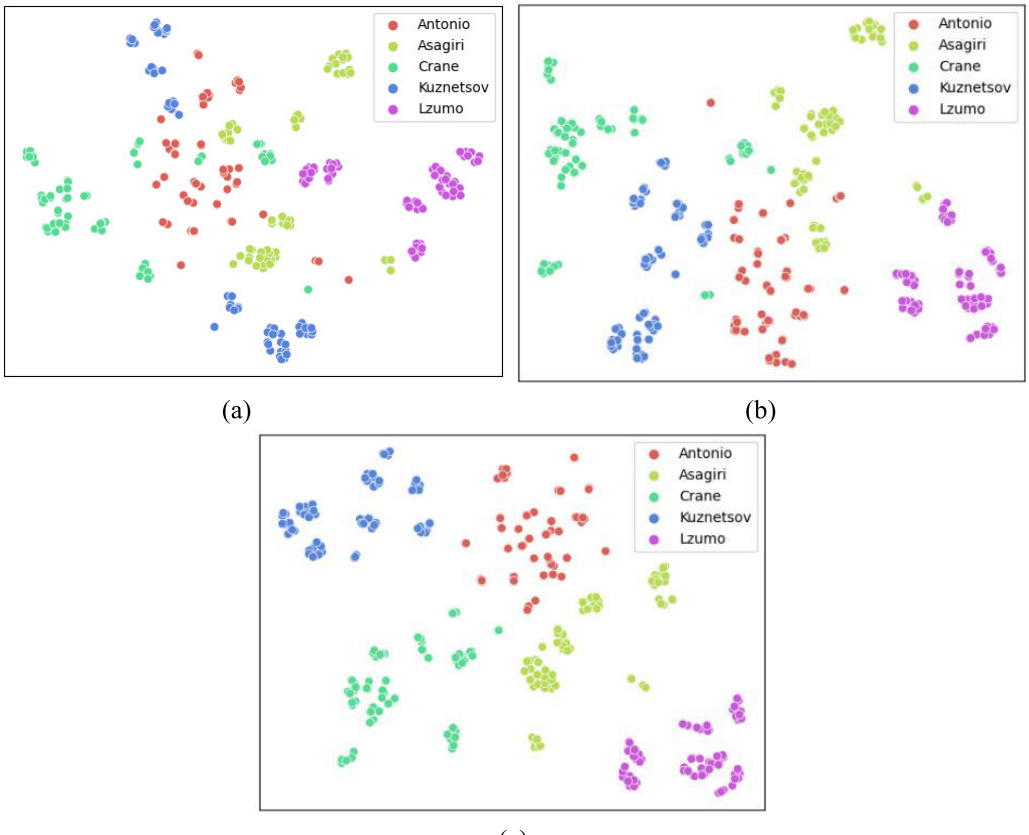



Fig. 7. Feature space of five evaluation sets from T-SNE. (a) Feature space with baseline network. (b) Feature space with LATA module. (c) Feature space with LATA module and bi-prototype module.

significantly facilitates ship classification, demonstrating the effectiveness of our attention module and the remarkable representation learning ability of the entire network.

*2) Visualization of Bi-Prototype Module:* We show that the bi-prototype module can force the network to keep the distance between the same-category samples smaller and

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different-category samples larger, thus improving the classification ability of the model. We visualized the feature space of the FGSCR dataset under a five-way-five-shot task using T-SNE [53] to validate our claims.

As shown in Fig. 7(a), the discriminability of the feature space formed by the baseline network is poor, and some categories are mixed. As shown in Fig. 7(b), with the addition of the LATA module, the separability between dissimilar categories is improved; however, some of the same categories are still relatively dispersed. As shown in Fig. 7(c), after adding both the LATA and bi-proto modules, the separability between the same categories is more compact, which significantly enhances the separability between different categories. This experiment proves the validity of our hypothesis.

### E. Limitations

Since the original support prototypes are not sufficiently discriminative to represent the differences in the category distribution of the queries, therefore, we use $L_{\text{pos}}$ and $L_{\text{rev}}$ generated by the bi-prototype module to assist the constraints, so as to generate more accurate prototypes. Among them, tradeoff hyperparameters ($\alpha$ and $\beta$) are used to control the constraint strength of $L_{\text{pos}}$ and $L_{\text{rev}}$. We try a variety of combinations of $\alpha$ and $\beta$ and find that the final performance differences are relatively slight when $\alpha$ and $\beta$ are in a specific interval [10, 0]. In this article, we report the optimal value selected by hand. In the future, we can consider using meta-learning method to make the model learn the ability to obtain $\alpha$ and $\beta$ in each episode, and the model can automatically get the optimal value in different sample sizes or different datasets, improving the experimental efficiency.

In addition, the LATA-BP-BDC is influenced by datasets. The FGSCR datasets are unevenly distributed, showing a long-tail effect, which leads to the failure to construct enough episodes in the training stage for categories with very few samples. Compared with evenly distributed CUB datasets, the performance difference is several percentage points. Therefore, in the future, we can consider to design a metric learning model with stronger generalization ability on the basis of the LATA-BP-BDC, and the model parameters obtained by training in a dataset with sufficient samples can be directly applied to test a dataset with few training samples (i.e., realizing cross-domain few-shot fine-grained image classification).

## V. Conclusion

The FSL plays an increasingly important role in the absence of available training samples. In this study, we propose LATA-BP-BDC for few-shot fine-grained ship classification. The proposed network is simple, effective, and scalable. For feature extraction, the LATA module is added to generate 3-D weights for the feature maps along the channel and spatial dimensions to filter irrelevant regions and the complex background and adaptively extract more representative features. For metric classification, a bi-prototype module can further calibrate the information differences between images by generating more accurate prototypes, thereby effectively reducing the influence of large intraclass appearance variations and small interclass appearance variations. Extensive experiments show that LATA-BP-BDC realizes the most advanced results on the FGSCR and CUB datasets and outperforms the state-of-the-art few-shot classification methods.

One should note that the Meta DeepBDC [37] is used as the baseline for this study. This baseline is originally devised to solve optical image classification problems. Therefore, the method developed in this study is suitable for addressing the issue of target classification in optical images. We propose the LATA-BP-BDC in view of the current shortcomings of insufficient feature extraction ability and image discrimination ability. Our direct goal is to solve the optical remote sensing image problem, and this study proves that the LATA-BP-BDC can be well used for ship classification in optical remote sensing images. In fact, because complex background interference, interclass similarity, and intraclass diversity are common problems in fine-grained image classification, LATA-BP-BDC is also useful for target classification in other images, such as natural images (cub dataset). The experiment also proves that the fine-grained classification of natural images also has good generalization. However, considering the personality of different types of images is necessary. Furthermore, understanding whether the LATA-BP-BDC can be used for other types of images remains open and may also be beyond the scope of this study. Such as SAR is the echo signal of radar scattering, which is closely related to the azimuth, viewing angle, target structure, and band. Therefore, how to transfer to the SAR task to further expand the fine-grained classification method proposed in this study is worthy of future research.

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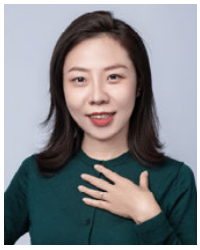
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